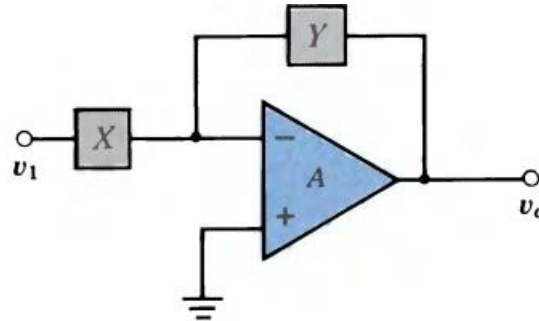
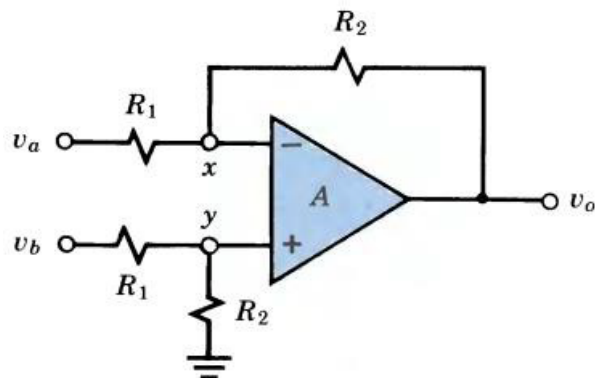


ELL 101: Introduction to Electrical Engineering

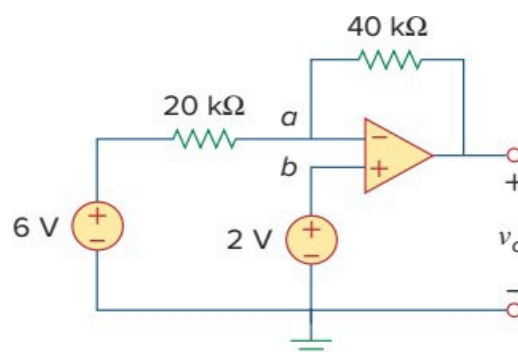
- For the circuit shown in the figure, where $X=L$ and $Y=R$, derive an expression for V_0 in terms of V_1 . What function is performed by the circuit?



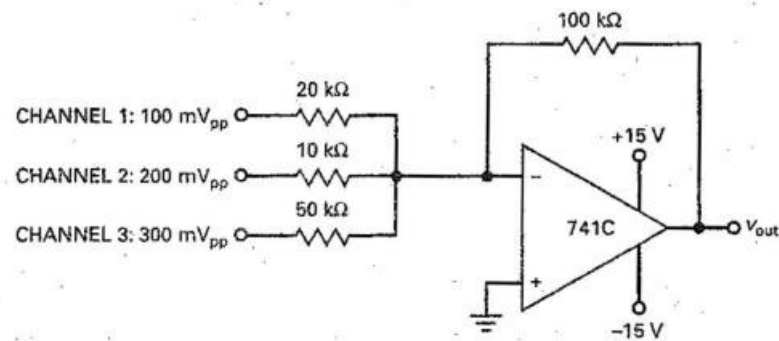
- Starting from basic principles and stating any assumptions:
 - Express v_x and v_y in terms of v_a , v_b and v_o in the given figure and derive expression for v_o .



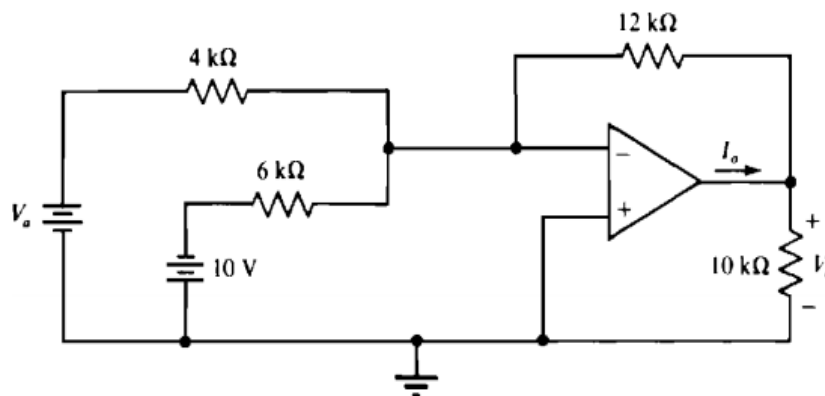
- Define the function of this circuit.
 - For $R_1 = R_2 = 50\text{k}\Omega$, $v_a = -3 + \sin\omega t$ and $v_b = \sin\omega t + 2 \sin\omega t$, predict v_o and explain the virtue of this circuit.
- A 741 op-amp with slew rate $SR = 0.5 \text{ V}/\mu\text{s}$ is used as part of a motor control system. If the highest reproducible frequency of AC signal is 3 kHz and the maximum output level is 12 V peak, does slewing ever occur?
 - Determine v_o in the op-amp circuit below.



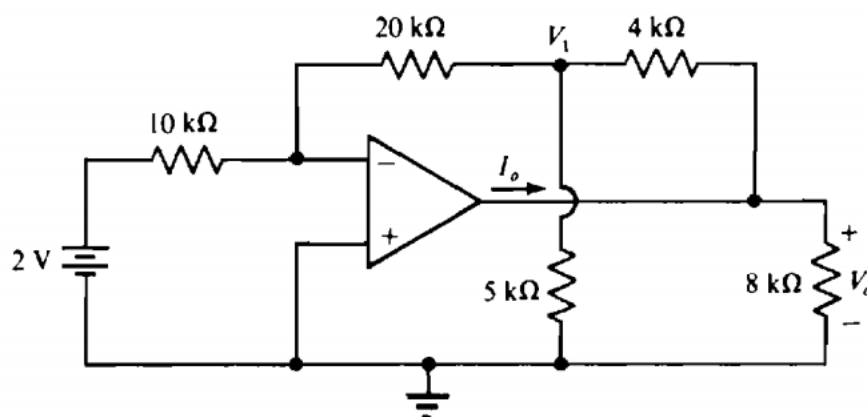
5. Three audio signals drive the summing amplifier as shown in figure. What is the output voltage?



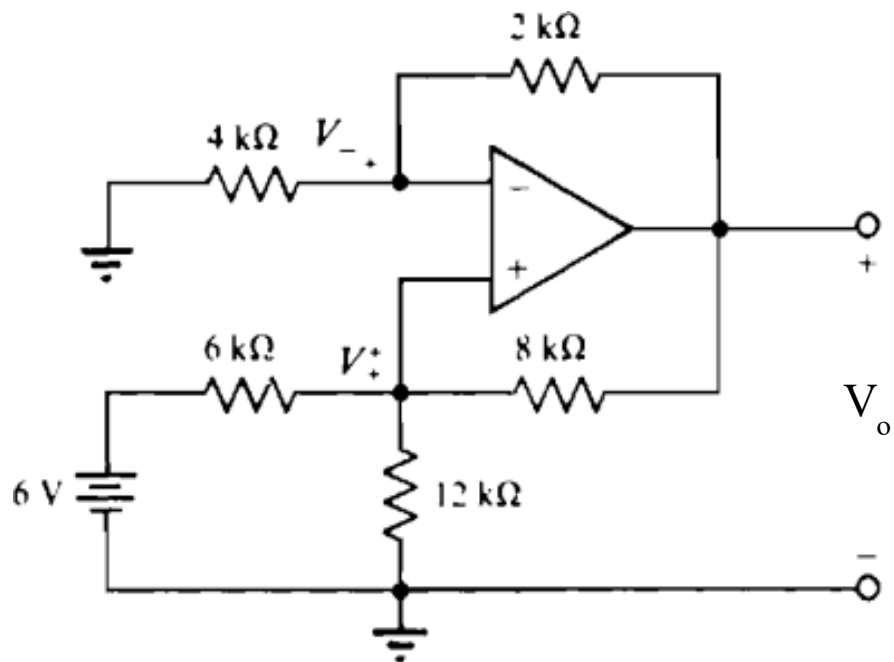
6. The input voltage to an op amp is a large voltage step. The output is an exponential waveform that changes to 0.25 V in 0.1 μ s. What is the slew rate of the op amp?
7. (a) For an input $V_a = 4$ V, find the output voltage V_o and current I_o
 (b) Determine the range of V_a for linear operation if the saturation voltage of op-amp is ± 12 V



8. Determine V_o and I_o in the circuit below



9. Find output voltage V_o for the circuit below.



10. If $v_1(t) = 10\cos(2t)$ mV and $v_2(t) = 0.5t$ mV below, find $v_o(t)$ for $t > 0$. Assume that voltage across the capacitor is zero at $t = 0$.

